

Gopalakrishna Maddipalli

GENERATIVE AI ENGINEER | DATA SCIENTIST

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PROFILE

Production-minded AI engineer with seven-plus years spanning construction operations, research, data science, and applied AI. Builds evidence-grounded RAG systems, multi-agent workflows, document intelligence, predictive models, and cloud APIs with explicit evaluation and human review.

CORE CAPABILITIES

Generative AI	LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, MCP, prompt engineering
Retrieval	Qdrant, Pinecone, Milvus, Chroma, FAISS, hybrid search, reranking, evaluation
Models & ML	Llama, Qwen, DeepSeek, PyTorch, scikit-learn, XGBoost, SHAP, LoRA / QLoRA
Engineering	Python, FastAPI, Docker, AWS, Azure Container Apps, Next.js, TypeScript, GitHub Actions
Data & documents	OCR, Textract, PaddleOCR, spaCy, PostgreSQL, PySpark, Athena, Power BI

EXPERIENCE

Applied Data Science & AI Systems

2021 - Present

Venusgeo Solutions | Tirupati Area, India

- Design and ship machine-learning, OCR, retrieval, and agentic systems for document-heavy construction and legal workflows.
- Build evidence-aware workflows with specialist agents, controlled correction, human review, and auditable system boundaries.
- Develop operational dashboards and APIs that connect work-order evidence, model outputs, and team decision-making.
- Partner with product and operations teams to frame use cases, prototype data-driven features, and evaluate system behavior.

Construction Research & Operations

2019 - 2021

Venusgeo Solutions | Tirupati Area, India

- Analyzed operational and customer information to support business, product, and workflow decisions.
- Developed construction-domain and compliance-process knowledge that now informs applied AI system design.
- Built the transition foundation from research and operations into data science and intelligent automation.

SELECTED AI SYSTEMS

Autonomous Work Order Intelligence & Operations Platform

Microsoft Foundry, Python, FastAPI, MCP, AWS, Azure Container Apps

- Evidence-aware multi-agent architecture for intake, research, discrepancy detection, controlled correction, and human escalation.

NTO Operations Copilot

Microsoft Foundry, MCP, FastAPI, Next.js, AWS Lambda, Azure App Service

- Evidence-grounded research coach guiding new users through notice selection, exceptions, and approved next actions.

Construction Payment Risk Prediction

Python, FastAPI, AWS, DataRobot, SHAP

- Human-reviewed operational prioritization with explainable risk signals for payment-protection workflows.

Construction Document Intelligence

Textract, PaddleOCR, spaCy, PostgreSQL, FastAPI

- Structured, reviewable extraction from complex legal PDFs to reduce manual document processing.

EDUCATION & SELECTED CREDENTIALS

Post Graduate Program in Artificial Intelligence and Machine Learning

McCombs School of Business, The University of Texas at Austin via Great Learning | Jan 2021

Bachelor of Technology - Mechanical Engineering

YSR Engineering College of YVU | 2014 - 2018

Agent Architect - Founderz AI & Business School | Sep 2026

Generative AI for Data Scientists - IBM / Coursera Specialization | Sep 2026

Generative AI for Data Analysts - IBM / Coursera Specialization | Sep 2026

Generative AI: Elevate your Software Development Career - IBM / Coursera | Sep 2026

Machine Learning Engineering for Production (MLOps) - Coursera Specialization | Nov 2021

Practical Data Science - Coursera Specialization | Sep 2021

Neural Networks and Deep Learning - Coursera | Sep 2020